

Worksheet Lab 2

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Abstract—This report presents a comprehensive analysis of training a ResNet-18 model on the CIFAR-10 dataset. The study emphasizes DL-Ops principles, including custom dataset orchestration, computational complexity profiling (FLOPs), and real-time telemetry using Weights & Biases (WandB). We analyze gradient stability and weight update dynamics over 30 epochs to validate model convergence and architectural efficiency.

Index Terms—ResNet-18, CIFAR-10, FLOPs, WandB, Gradient Flow, DL-Ops

I. INTRODUCTION

Modern Deep Learning Operations (DL-Ops) require rigorous monitoring of internal model states. This lab explores the training of a ResNet-18 architecture, specifically focusing on the "pulse" of the network through gradient flow and weight update telemetry. By utilizing WandB, we visualize how skip connections influence backpropagation and how computational costs scale with input dimensions.

II. EXPERIMENTAL SETUP

A. Custom Dataset and Dataloader

A custom-made PyTorch `Dataset` class was implemented to wrap the CIFAR-10 data. This ensures a modular pipeline where data augmentation (Random Cropping, Horizontal Flipping) and normalization ($mean = [0.4914, 0.4822, 0.4465]$) are decoupled from the core training logic.

B. Architecture and Optimizer

The model utilized is ResNet-18, adapted for 32×32 input images.

- **Optimizer:** SGD with Momentum (0.9) and Weight Decay ($5e - 4$).
- **Scheduler:** Cosine Annealing LR (starting at 0.1).
- **Epochs:** 30.

III. COMPUTATIONAL PROFILING (FLOPs)

Using the `thop` profiling library, the computational footprint of the model was measured. For a single forward pass of a $32 \times 32 \times 3$ image, the results are:

- **Total FLOPs:** 37.22 Million
- **Total Parameters:** 11.17 Million

The relatively low FLOP count compared to standard ImageNet-scale ResNet-18 models is due to the smaller spatial resolution of CIFAR-10, significantly reducing the complexity of the initial convolutional layers.

IV. RESULTS AND TELEMETRY ANALYSIS

A. Loss Convergence

The training session (Run ID: kbjed4d2) showed consistent convergence. The loss dropped from **2.1622** in Epoch 1 to **0.2668** by Epoch 30.



Fig. 1. Training Loss profile over 30 epochs showing smooth decay using Cosine Annealing.

B. Gradient Flow Analysis

WandB telemetry captured the average gradient magnitudes across layers. Table I summarizes the final state of the gradients recorded at the end of training.

TABLE I
LAYER-WISE GRADIENT MEAN (EPOCH 30 - RUN KBJED4D2)

Layer Name	Mean Gradient Magnitude
bn1.weight	0.02142
conv1.weight	0.00629
layer1.0.bn1.weight	0.01398
layer1.0.conv1.weight	0.00178
layer1.1.bn1.weight	0.01934
fc.weight	0.00155

C. Observations on Weight Updates

The weight update flow closely followed the Cosine Annealing learning rate schedule. High initial update magnitudes indicated rapid feature discovery, while the convergence in epochs 25–30 demonstrated the model fine-tuning its parameters as the learning rate approached its minimum.

V. DISCUSSION AND CONCLUSION

The experiment confirms that ResNet-18 is highly optimized for CIFAR-10 tasks. The constant gradient signal across layers proves that residual connections successfully mitigate the vanishing gradient problem. The computational cost of 37.22M FLOPs makes this model suitable for real-time DL-Ops applications on edge devices. All telemetry data and experimental logs are hosted on Weights & Biases for reproducibility.

REPRODUCIBILITY AND LINKS

- **WandB Project:** Project Dashboard
- **GitHub Repository:** Source Code